National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Computer Networks	Course Code:	CS3001
Program:	BS(DATA Science)	Semester:	Spring 25
Duration	20 minutes	Total Marks:	15
Paper Date:	8 May -2025	Section:	8A
Exam Type:	Quiz 5 – Chapter 5	Page(s):	2

Name: solution **Roll No.** **Section:**

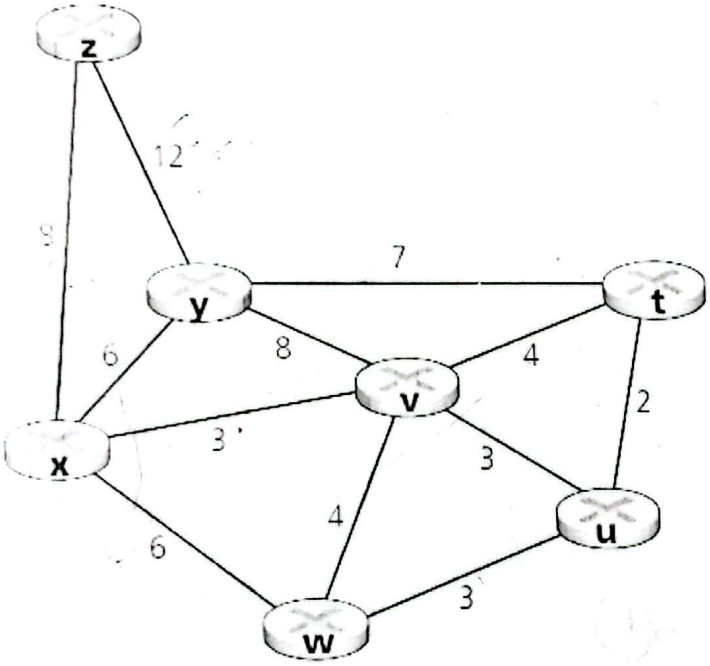
Question 01: Answer the multiple-choice questions by choosing one option.

1. Inter-AS routing is_____ while Intra-AS routing is_____?
   a. Within an AS, between different ASes
   b. Outside an AS, between different ASes
   c. Between different ASes, within an AS
   d. None of the above

2. RIP uses_____ while OSPF uses_____:
   a. Distance vector, Link State
   b. Link State, Distance Vector
   c. Both Link State
   d. None of the above

3. If direct link cost between node w and node z is infinity this indicates:
   a. There is no path between w and z
   b. There is a direct link between w and z
   c. There is no direct link between w and z
   d. There is a path between w and z, but another path is better

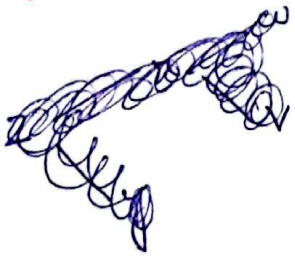
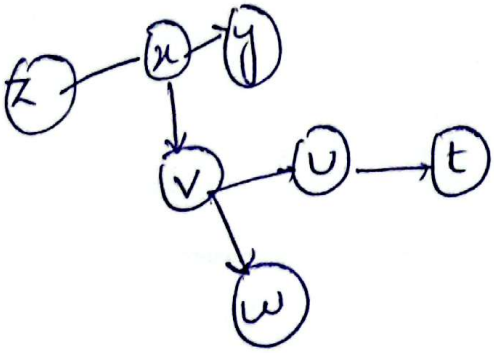
Q2. Run Dijkstra's algorithm with source node Z. Create table for showing all iterations, forwarding table and minimum spanning tree.



D

Steps	N'	D(t),p(t)	D(u),p(u)	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)
0	z	∞	∞	∞	∞	8,z	12,z
1	zx	∞	∞	11,x	14,x	8,z	12,z
2	zxv	15,v	14,v	11,x	14,x	8,z	12,z
3	zxvy	15,v	14,v	11,x	14,x	8,z	12,z
4	zxvyu	15,v	14,v	11,x	14,x	8,z	12,z
5	zxvyuw	15,v	14,v	11,x	14,x	8,z	12,z
6	zxvyuwt	15,v	14,v	11,x	14,x	8,z	12,z

~~forwarding table~~



Spanning Tree:-